

AWS Cloud Infrastructure Deployment Project

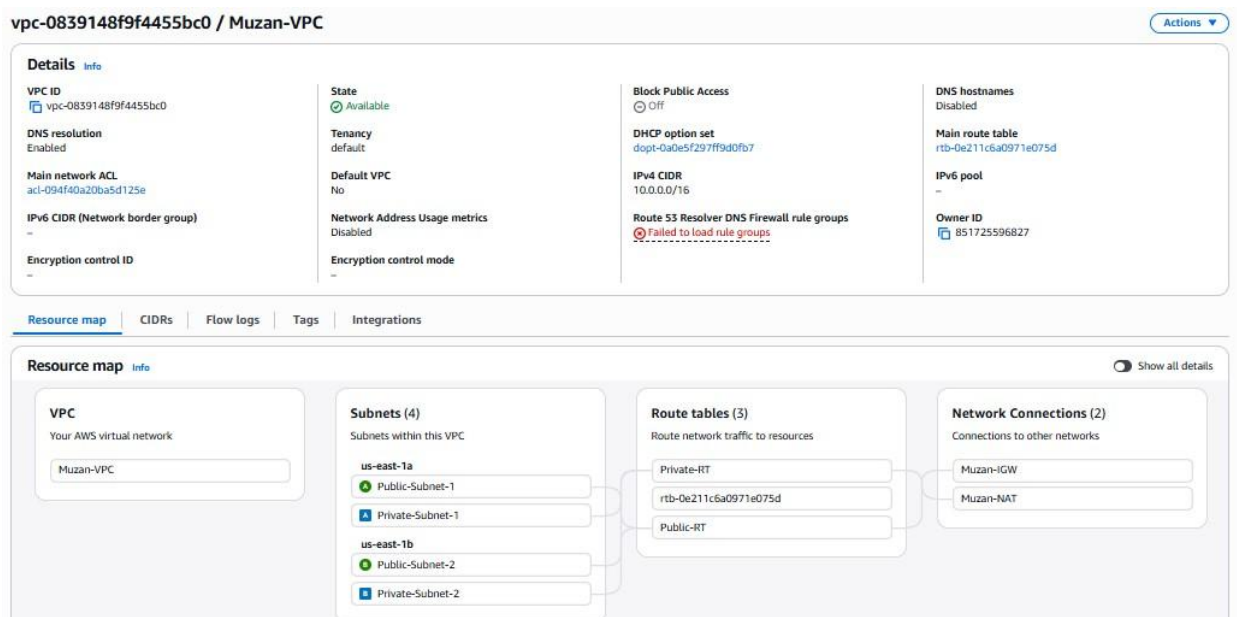
Done by: Muzan Abbas

SECTION 1 – VPC & NETWORKING

Explanation:

I created a Virtual Private Cloud (VPC) using the CIDR block 10.0.0.0/16 as required. Inside the VPC, I set up four subnets across two Availability Zones to make the system more reliable. These include two public subnets (10.0.0.0/24 and 10.0.2.0/24) and two private subnets (10.0.1.0/24 and 10.0.3.0/24), distributed across different Availability Zones. I attached an Internet Gateway to the VPC so that resources in the public subnets can access the internet. I also created a NAT Gateway in Public Subnet 1 to allow instances in the private subnets to access the internet without being directly exposed. The route tables were configured so that public subnets send traffic to the Internet Gateway, while private subnets send traffic to the NAT Gateway. This setup keeps the architecture organized and ensures that private resources remain secure while still having outbound access.

1) VPC Creation



2) 4 Subnets with their CIDRs + Availability Zones

Name	Subnet ID	State	VPC	Block Public Access	IPv4 CIDR	IPv6 CIDR	IPv4 CIDR association ID	Available IPv4 addresses	Availability Zone	Network border group	Route table	Notes
Public-Subnet-1	subnet-0a0f7c138603d85	Available	vpc-0839148f9f4455bc0	Off	172.31.48.0/24	-	-	4091	us-east-1 us-east-1a	us-east-1	rtb-0e211c6a0971e075d	Public
Public-Subnet-2	subnet-0a0f7c138603d85	Available	vpc-0839148f9f4455bc0	Off	10.0.0.0/24	-	-	248	us-east-1 us-east-1b	us-east-1	rtb-0e211c6a0971e075d	Public
Private-Subnet-1	subnet-0a0f7c138603d85	Available	vpc-0839148f9f4455bc0	Off	172.31.48.0/24	-	-	4091	us-east-1 us-east-1c	us-east-1	rtb-0e211c6a0971e075d	Private
Private-Subnet-2	subnet-0a0f7c138603d85	Available	vpc-0839148f9f4455bc0	Off	10.0.0.0/24	-	-	248	us-east-1 us-east-1d	us-east-1	rtb-0e211c6a0971e075d	Private

3) CIDR

vpc-0839148f9f4455bc0 / Muzan-VPC

Details Info

VPC ID
vpc-0839148f9f4455bc0

DNS resolution
Enabled

Main network ACL
acl-094f40a20ba5d125e

IPv6 CIDR (Network border group)
-

Encryption control ID
-

State
Available

Tenancy
default

Default VPC
No

Network Address Usage metrics
Disabled

Encryption control mode
-

Block Public Access
Off

DHCP option set
dopt-0a0e5f297ff9d0fb7

IPv4 CIDR
10.0.0.0/16

Route 53 Resolver DNS Firewall rule groups
Failed to load rule groups

DNS hostnames
Disabled

Main route table
rtb-0e211c6a0971e075d

IPv6 pool
-

Owner ID
851725596827

Resource map

CIDRs

Flow logs

Tags

Integrations

IPv4 CIDRs Info

Edit CIDRs

Address family

▲ | CIDR

IPv4

10.0.0.0/16



4) IGW attached to Muzan VPC

igw-04874d9d2decf59ab / Muzan-IGW

Actions

Details

Internet gateway ID
igw-04874d9d2decf59ab

State
Attached

VPC ID
vpc-0839148f9f4455bc0 | Muzan-VPC

Owner
851725596827

Tags (1)

Manage tags

Search tags

Key	Value
Name	Muzan-IGW

5) NAT Gateway

nat-0f63364fdb5575043 / Muzan-NAT

Actions

Details

NAT gateway ID
nat-0f63364fdb5575043

NAT gateway ARN
arn:aws:ec2:us-east-1:851725596827:natgateway/nat-0f63364fdb5575043

VPC
vpc-0839148f9f4455bc0 | Muzan-VPC

Connectivity type
Public

Primary public IPv4 address
44.212.175.145

Subnet
subnet-081f91c9dde3dc67d | Public-Subnet-1

State
Available

Primary private IPv4 address
10.0.0.197

Created
Friday, April 3, 2026 at 15:59:02 GMT+3

State message
-

Primary network interface ID
eni-Oad1364c5d83a99b7

Deleted
-

Secondary IPv4 addresses

Monitoring

Tags

Search

Private IPv4 address	Allocation ID	Association ID	Public IPv4 address	Network interface ID	Status
Secondary IPv4 addresses are not available for this nat gateway.					

Edit secondary IPv4 address associations

6) Route Tables (1 Public and 1 Private)

Public Route Table with 0.0.0.0/0  IGW

rtb-08dd66f94dbcde1fd / Public-RT

Actions

Details

Route table ID
rtb-08dd66f94dbcde1fd

VPC
vpc-0839148f9f4455bc0 | Muzan-VPC

Main
No

Owner ID
851725596827

Explicit subnet associations
2 subnets

Edge associations
-

Routes

Subnet associations

Edge associations

Route propagation

Tags

Filter routes

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	igw-04874d9d2decf59ab	Active	No	Create Route
10.0.0.0/16	local	Active	No	Create Route Table

Both

Edit routes

Associated with 2 public subnets

rtb-08dd66f94dbcd1fd / Public-RT

Actions

Details

Info

Route table ID

rtb-08dd66f94dbcd1fd

VPC

vpc-0839148f9f4455bc0 | Muzan-VPC

Main

No

Owner ID

851725596827

Explicit subnet associations

2 subnets

Edge associations

-

Routes

Subnet associations

Edge associations

Route propagation

Tags

Explicit subnet associations (2)

Edit subnet associations

Find subnet association

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR
Public-Subnet-2	subnet-0048d0dd8079dd5c	10.0.2.0/24	-
Public-Subnet-1	subnet-081f91c9dde3dc67d	10.0.0.0/24	-

Subnets without explicit associations (0)

Edit subnet associations

The following subnets have not been explicitly associated with any route tables and are therefore associated with the main route table:

Find subnet association

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR
------	-----------	-----------	-----------

No subnets without explicit associations
All your subnets are associated with a route table.

Private Route Table 0.0.0.0/0 NAT Gateway

rtb-067870f5f061ff588 / Private-RT

Actions

Details

Info

Route table ID

rtb-067870f5f061ff588

VPC

vpc-0839148f9f4455bc0 | Muzan-VPC

Main

No

Owner ID

851725596827

Explicit subnet associations

2 subnets

Edge associations

-

Routes

Subnet associations

Edge associations

Route propagation

Tags

Routes (2)

Both

Edit routes

Filter routes

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	nat-0f63364fdb5575043	Active	No	Create Route
10.0.0.0/16	local	Active	No	Create Route Table

Associated with 2 Private subnets

rtb-067870f5f061ff588 / Private-RT

Actions

Details

Route table ID

rtb-067870f5f061ff588

VPC

vpc-0839148f9f4455bc0 | Muzan-VPC

Main

No

Owner ID

851725596827

Explicit subnet associations

2 subnets

Edge associations

-

Routes

Subnet associations

Edge associations

Route propagation

Tags

Explicit subnet associations (2)

Find subnet association

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR
Private-Subnet-2	subnet-0186a826ab12fac46	10.0.3.0/24	-
Private-Subnet-1	subnet-04106b5173f16579a	10.0.1.0/24	-

Subnets without explicit associations (0)

The following subnets have not been explicitly associated with any route tables and are therefore associated with the main route table:

Find subnet association

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR
------	-----------	-----------	-----------

No subnets without explicit associations
All your subnets are associated with a route table.

FULL VPC RESOURCE MAP FOR DEMO

Resource map

CIDRs

Flow logs

Tags

Integrations

Resource map

Info

Show all details

VPC

Your AWS virtual network

Muzan-VPC

Subnets (4)

Subnets within this VPC

us-east-1a

Public-Subnet-1

Private-Subnet-1

us-east-1b

Public-Subnet-2

Private-Subnet-2

Route tables (3)

Route network traffic to resources

Private-RT

rtb-0e211c6a0971e075d

Public-RT

Network Connections (2)

Connections to other networks

Muzan-IGW

Muzan-NAT

(0.0.0.0/0), so users can access the application through the internet. The Web Tier Security Group only allows HTTP traffic on port 80 from the load balancer, which means the web servers are not directly exposed to the internet. The Database Security Group allows MySQL traffic on port 3306 only from the web tier, ensuring that the database is fully private. This setup follows a layered security approach and makes sure each component only communicates with what it needs to.

1) Load Balancer SG (sourced from 0.0.0.0/0 for both protocols)

sg-0e1e8fb04120befcd - ALB-SG Actions

Details

Security group name ALB-SG	Security group ID sg-0e1e8fb04120befcd	Description Allow HTTP/HTTPS from internet	VPC ID vpc-0839148f9f4455bc0
Owner 851725596827	Inbound rules count 2 Permission entries	Outbound rules count 1 Permission entry	

[Inbound rules](#) | [Outbound rules](#) | [Sharing](#) | [VPC associations](#) | [Related resources - new](#) | [Tags](#)

Inbound rules (2) Manage tags Edit inbound rules

Search

Name	Security group rule ID	IP version	Type	Protocol	Port range	Source	Description
-	sgr-005108d841658c8eb	IPv4	HTTP	TCP	80	0.0.0.0/0	-
-	sgr-08fa520c1bb8e4770	IPv4	HTTPS	TCP	443	0.0.0.0/0	-

2) Web Tier SG (sourced from load balancer sg ALB)

sg-00b37e4f6afead3c7 - WebTier-SG Actions

Details

Security group name WebTier-SG	Security group ID sg-00b37e4f6afead3c7	Description Allows inbound HTTP traffic only from the Application Load Balancer security group, preventing direct internet access to web servers in private subnets	VPC ID vpc-0839148f9f4455bc0
Owner 851725596827	Inbound rules count 1 Permission entry	Outbound rules count 1 Permission entry	

[Inbound rules](#) | [Outbound rules](#) | [Sharing](#) | [VPC associations](#) | [Related resources - new](#) | [Tags](#)

Inbound rules (1) Manage tags Edit inbound rules

Search

Name	Security group rule ID	IP version	Type	Protocol	Port range	Source	Description
-	sgr-048b3fdb3d881ab2	-	HTTP	TCP	80	sg-0e1e8fb04120befcd...	-

sg-0e1e8fb04120befcd - ALB-SG Actions

Details

Security group name ALB-SG	Security group ID sg-0e1e8fb04120befcd	Description Allow HTTP/HTTPS from internet	VPC ID vpc-0839148f9f4455bc0
Owner 851725596827	Inbound rules count 2 Permission entries	Outbound rules count 1 Permission entry	

[Inbound rules](#) | [Outbound rules](#) | [Sharing](#) | [VPC associations](#) | [Related resources - new](#) | [Tags](#)

Inbound rules (2) Manage tags Edit inbound rules

Search

Name	Security group rule ID	IP version	Type	Protocol	Port range	Source	Description
-	sgr-005108d841658c8eb	IPv4	HTTP	TCP	80	0.0.0.0/0	-
-	sgr-08fa520c1bb8e4770	IPv4	HTTPS	TCP	443	0.0.0.0/0	-

3) Database SG (sourced from web tier sg)

sg-0bc6f89f41e8a5e23 - Database-SG Actions

Details

Security group name Database-SG	Security group ID sg-0bc6f89f41e8a5e23	Description Allows inbound MySQL traffic on port 3306 only from the WebTier-SG, ensuring the database is inaccessible from the internet or load balancer directly	VPC ID vpc-0839148f9f4455bc0
Owner 851725596827	Inbound rules count 1 Permission entry	Outbound rules count 1 Permission entry	

Inbound rules | Outbound rules | Sharing | VPC associations | Related resources - new | Tags

Inbound rules (1) Manage tags Edit inbound rules

Name	Security group rule ID	IP version	Type	Protocol	Port range	Source	Description
-	sg-08779f51327e33291	-	MYSQL/Aurora	TCP	3306	sg-00b37e4f6afead3c7...	Allow MySQL from We...

sg-00b37e4f6afead3c7 - WebTier-SG Actions

Details

Security group name WebTier-SG	Security group ID sg-00b37e4f6afead3c7	Description Allows inbound HTTP traffic only from the Application Load Balancer security group, preventing direct internet access to web servers in private subnets	VPC ID vpc-0839148f9f4455bc0
Owner 851725596827	Inbound rules count 1 Permission entry	Outbound rules count 1 Permission entry	

Inbound rules | Outbound rules | Sharing | VPC associations | Related resources - new | Tags

Inbound rules (1) Manage tags Edit inbound rules

Name	Security group rule ID	IP version	Type	Protocol	Port range	Source	Description
-	sg-048b3fdb3d881ab2	-	HTTP	TCP	80	sg-0e1e8fb04120befcd...	-

SECTION 3 - RDS Database

Explanation:

I set up an Amazon RDS MySQL database in the private subnets to keep it secure. The database is configured with Multi-AZ enabled, which improves availability by creating a standby instance in another Availability Zone. It is not publicly accessible, and it uses the Database Security Group that only allows connections from the web tier on port 3306. This ensures that the database can only be accessed by the application servers and not from the internet.

- 1) DB engine: MYSQL, Status: Available, Multi-AZ: Enabled

Databases (1)

Filter by databases

DB identifier	Status	Role	Engine	Upgrade rollout ...	Region ...	Size	Recommendations	CPU	Current
mucan-db	Available	Instance	MySQL Community	SECOND	us-east-1b	db.t3.medium	3 Informational	5.19%	

Databases (1)

Filter by databases

Role	Engine	Upgrade rollout ...	Region ...	Size	Recommendations	CPU	Current...	Mainte...	VPC	Multi-AZ
Instance	MySQL Community	SECOND	us-east-1b	db.t3.medium	3 Informational	5.19%	0 Contr	none	vpc-0839148f9f4455bc0	Yes

2) Private Subnets

private db

Subnet group details

VPC ID
vpc-0839148f9f4455bc0

ARN
arn:aws:rds:us-east-1:851725596827:subgrp:private-db

Supported network types
IPv4

Description
Subnet group with private subnets for RDS

Subnets (2)

Availability zone	Subnet name	Subnet ID	CIDR block
us-east-1b	Private-Subnet-2	subnet-0186a826ab12fac45	10.0.3.0/24
us-east-1a	Private-Subnet-1	subnet-04106b5173f16579a	10.0.1.0/24

Tags (0)

Filter by Tag key

Tags

You don't have any tags associated with this resource.

Manage tags

Your VPCs

VPCs | VPC encryption controls

Your VPCs (1) info

Find VPCs by attribute or tag

VPC-0839148f9f4455bc0

Name	VPC ID	State	Encryption c...	Encryption control...	Block Public...	IPv4 CIDR	IPv6 CIDR	DHCP option set	Main route table	Main network ACL	Tenancy	Default VPC	Owner ID
Private VPC	vpc-0839148f9f4455bc0	Available	-	-	Off	10.0.0.0/16	-	default-2017-09-27	rtb-de2113bae977a679d	acl-c9af6a00ba0d125a	default	No	851725596827

3) Security Groups (Database SG with inbound traffic from port 3306)

Security group rules (2)

Security group	Type	Rule
Database-SG sg-0bc6f89f41e8a5e23	EC2 Security Group - Inbound	sg-08c37e4f6afead3c7
Database-SG sg-0bc6f89f41e8a5e23	CDN/VP - Outbound	0.0.0.0/0

sg-0bc6f89f41e8a5e23 - Database-SG

Details

Security group name
Database-SG

Owner
851725596827

Security group ID
sg-0bc6f89f41e8a5e23

Inbound rules count
1 Permission entry

Description
Allows inbound MySQL traffic on port 3306 only from the WebTier-SG, ensuring the database is inaccessible from the internet or load balancer directly

Outbound rules count
1 Permission entry

VPC ID
[vpc-0839148f9f4455b0d](#)

[Inbound rules](#) | [Outbound rules](#) | [Sharing](#) | [VPC associations](#) | [Related resources - new](#) | [Tags](#)

Inbound rules (1)

Name	Security group rule ID	IP version	Type	Protocol	Port range	Source	Description
-	sg-08779f51327e33291	-	MySQL/Aurora	TCP	3306	sg-00b37e4f6afead3c7 ...	Allow MySQL from We...

SECTION 4 – Load Balancer + Web Tier

Explanation:

I created an internet-facing Application Load Balancer and placed it in the two public subnets so it can receive traffic from users. The load balancer listens on port 80 and forwards requests to a target group. The target group is configured to route traffic to EC2 instances running in the private subnets. These instances are launched using a launch template with the required t2.micro instance type. Since the web servers are in private subnets, they are not directly accessible from the internet, and all traffic must go through the load balancer. This improves both security and performance.

1) Load Balancer type: application, Scheme: internet-facing, Public subnets

Load balancers (1) [What's new?](#)

Elastic Load Balancing scales your load balancer capacity automatically in response to changes in incoming traffic.

Filter load balancers

☐	Name	State	Type	Scheme	IP address type	VPC ID	Availability Zones	Security groups	DNS name	ARN	Date created
☐	Muzan-ALB	Active	application	Internet-facing	IPv4	vpc-0839148f9f4455bc0	2 Availability Zones	sg-0e1e8fb04120befcd	Muzan-ALB-1389468852.us-east-1.elb.amazonaws.com	ARN	April 5, 2026, 17:26 (UTC+03:00)

2) Listener

Muzan-ALB

Details

Load balancer type Application	Status Active	VPC vpc-0839148f9f4455bc0	Load balancer IP address type IPv4
Scheme Internet-facing	Hosted zone Z35SXDOTRQ7X7K	Availability Zones subnet-0048d0dd8079dd5c us-east-1b (use1-az1) subnet-081f91c9dde3dc67d us-east-1a (use1-az6)	Date created April 5, 2026, 17:26 (UTC+03:00)
Load balancer ARN arn:aws:elasticloadbalancing:us-east-1:851725596827:loadbalancer/app/Muzan-ALB/af3fa2760e34a3d0		DNS name Muzan-ALB-1389468852.us-east-1.elb.amazonaws.com (A Record)	

Listeners and rules | Network mapping | Resource map | Security | Monitoring | Integrations | Attributes | Capacity | Tags

Listeners and rules (1) [Info](#)

A listener checks for connection requests on its configured protocol and port. Traffic received by the listener is routed according to the default action and any additional rules.

Filter listeners

☐	Protocol:Port	Default action	Rules	ARN	Security policy	Default SSL/TLS certificate	mTLS	Trust store
☐	HTTP:80	Forward to target group WebTier-TG (100%) Target group stickiness: Off	1 rule	ARN	Not applicable	Not applicable	Not applicable	Not applicable

3) Target Group port 80 and instance type

WebTier-TG

Actions

Details

arn:aws:elasticloadbalancing:us-east-1:851725596827:targetgroup/WebTier-TG/08bab355c08c7122

Target type

Instance

Protocol : Port

HTTP: 80

Protocol version

HTTP1

VPC

vpc-0839148f9f4455bc0

IP address type

IPv4

Load balancer

Muzan-ALB

2

Total targets

2

Healthy

0

Unhealthy

0

Unused

0

Initial

0

Draining

0

Anomalous

▼ Distribution of targets by Availability Zone (AZ)

Select values in this table to see corresponding filters applied to the Registered targets table below.

Last fetched seconds ago

Zone	Total targets	Healthy	Unhealthy	Unused	Initial	Drain
us-east-1a (use1-az6)	1	1	0	0	0	0
us-east-1b (use1-az1)	1	1	0	0	0	0

Targets

Monitoring

Health checks

Attributes

Tags

Registered targets (2)

Info

Anomaly mitigation: Not applicable

Deregister

Register targets

Target groups route requests to individual registered targets using the protocol and port number specified. Health checks are performed on all registered targets according to the target group's health check settings. Anomaly detection is automatically applied to HTTP/HTTPS target groups with at least 3 healthy targets.

Filter targets

Instance ID	Name	Port	Zone	Health status	Health status details	Admini...	Overrid...	Launch...	Anomaly detectio...
i-0b9d99f42d3d1c46f	-	80	us-east-1b (us...	Healthy	-	No override.	No overrid...	April 6, 20...	Normal
i-01ef999f082c51853	-	80	us-east-1a (use...	Healthy	-	No override.	No overrid...	April 6, 20...	Normal

4) Registered Targets

WebTier-TG

Actions

Details

arn:aws:elasticloadbalancing:us-east-1:851725596827:targetgroup/WebTier-TG/08bab355c08c7122

Target type

Instance

Protocol : Port

HTTP: 80

Protocol version

HTTP1

VPC

vpc-0839148f9f4455bc0

IP address type

IPv4

Load balancer

Muzan-ALB

2

Total targets

2

Healthy

0

Unhealthy

0

Unused

0

Initial

0

Draining

0

Anomalous

▼ Distribution of targets by Availability Zone (AZ)

Select values in this table to see corresponding filters applied to the Registered targets table below.

Last fetched 2 minutes ago

Zone	Total targets	Healthy	Unhealthy	Unused	Initial	Drain
us-east-1a (use1-az6)	1	1	0	0	0	0
us-east-1b (use1-az1)	1	1	0	0	0	0

Targets

Monitoring

Health checks

Attributes

Tags

Registered targets (2)

Info

Anomaly mitigation: Not applicable

Deregister

Register targets

Target groups route requests to individual registered targets using the protocol and port number specified. Health checks are performed on all registered targets according to the target group's health check settings. Anomaly detection is automatically applied to HTTP/HTTPS target groups with at least 3 healthy targets.

Filter targets

Instance ID	Name	Port	Zone	Health status	Health status details	Admini...	Overrid...	Launch...	Anomaly detectio...
i-0b9d99f42d3d1c46f	-	80	us-east-1b (us...	Healthy	-	No override.	No overrid...	April 6, 20...	Normal
i-01ef999f082c51853	-	80	us-east-1a (use...	Healthy	-	No override.	No overrid...	April 6, 20...	Normal

Instance 1

Instance summary for i-0b9d99f42d3d1c46f [Info](#)

Updated less than a minute ago

[Connect](#)[Instance state ▼](#)[Actions ▼](#)

Instance ID
[i-0b9d99f42d3d1c46f](#)

IPv6 address
-

Hostname type
IP name: ip-10-0-3-171.ec2.internal

Answer private resource DNS name
-

Auto-assigned IP address
-

IAM role
-

IMDSv2
Required

Operator
-

Public IPv4 address
-

Instance state
 Running

Private IP DNS name (IPv4 only)
[ip-10-0-3-171.ec2.internal](#)

Instance type
t2.micro

VPC ID
[vpc-0839148f9f4455bc0 \(Muzan-VPC\)](#)

Subnet ID
[subnet-0186a826ab12fac46 \(Private-Subnet-2\)](#)

Instance ARN
[arn:aws:ec2:us-east-1:851725596827:instance/i-0b9d99f42d3d1c46f](#)

Private IPv4 addresses
[10.0.3.171](#)

Public DNS
-

Elastic IP addresses
-

AWS Compute Optimizer finding
[Opt-in to AWS Compute Optimizer for recommendations. | Learn more](#)

Auto Scaling Group name
[Muzan-ASG](#)

Managed
false

[Details](#) | [Status and alarms](#) | [Monitoring](#) | [Security](#) | [Networking](#) | [Storage](#) | [Tags](#)

▼ Instance details [Info](#)

AMI ID
[ami-01b14b7ad41e17ba4](#)

AMI name
[al2023-ami-2023.10.20260330.0-kernel-6.1-x86_64](#)

Stop protection
Disabled

Instance reboot migration
Default (On)

Stop-hibernate behavior

Monitoring
disabled

Allowed image
-

Launch time
[Mon Apr 06 2026 20:46:00 GMT+0300 \(Arabian Standard Time\) \(about 2 hours\)](#)

Instance auto-recovery
Default

AMI Launch index

Platform details
[Linux/UNIX](#)

Termination protection
Disabled

AMI location
[amazon/al2023-ami-2023.10.20260330.0-kernel-6.1-x86_64](#)

Lifecycle
normal

Key pair assigned at launch

Instance 2

Instance summary for i-01ef999f082c51853 [Info](#)

Updated 1 minute ago

[Connect](#)[Instance state ▼](#)[Actions ▼](#)

Instance ID
[i-01ef999f082c51853](#)

IPv6 address
-

Hostname type
IP name: ip-10-0-1-237.ec2.internal

Answer private resource DNS name
-

Auto-assigned IP address
-

IAM role
-

IMDSv2
Required

Operator
-

Public IPv4 address
-

Instance state
 Running

Private IP DNS name (IPv4 only)
[ip-10-0-1-237.ec2.internal](#)

Instance type
t2.micro

VPC ID
[vpc-0839148f9f4455bc0 \(Muzan-VPC\)](#)

Subnet ID
[subnet-04106b5173f16579a \(Private-Subnet-1\)](#)

Instance ARN
[arn:aws:ec2:us-east-1:851725596827:instance/i-01ef999f082c51853](#)

Private IPv4 addresses
[10.0.1.237](#)

Public DNS
-

Elastic IP addresses
-

AWS Compute Optimizer finding
[Opt-in to AWS Compute Optimizer for recommendations. | Learn more](#)

Auto Scaling Group name
[Muzan-ASG](#)

Managed
false

[Details](#) | [Status and alarms](#) | [Monitoring](#) | [Security](#) | [Networking](#) | [Storage](#) | [Tags](#)

▼ Instance details [Info](#)

AMI ID
[ami-01b14b7ad41e17ba4](#)

AMI name
[al2023-ami-2023.10.20260330.0-kernel-6.1-x86_64](#)

Stop protection
Disabled

Instance reboot migration
Default (On)

Stop-hibernate behavior

Monitoring
disabled

Allowed image
-

Launch time
[Mon Apr 06 2026 20:43:58 GMT+0300 \(Arabian Standard Time\) \(about 2 hours\)](#)

Instance auto-recovery
Default

AMI Launch index

Platform details
[Linux/UNIX](#)

Termination protection
Disabled

AMI location
[amazon/al2023-ami-2023.10.20260330.0-kernel-6.1-x86_64](#)

Lifecycle
normal

Key pair assigned at launch

Section 5 – Auto Scaling

Explanation:

I configured an Auto Scaling Group to manage the web tier instances across the two private subnets. The desired capacity is set to 2 instances, with a minimum of 2 and a maximum of 6, as required. I also set up a scaling policy based on CPU utilization, with a target of 60 percent. CloudWatch automatically monitors the CPU usage and triggers scaling when needed. This allows the system to handle increases in traffic while maintaining performance and avoiding unnecessary resource usage.

1) Launch Template

Instance type: t2.micro

WebTier-LT (lt-0ddafcd07656d5c5) Actions ▼ Delete template

Launch template details

Launch template ID lt-0ddafcd07656d5c5	Launch template name WebTier-LT	Default version 1	Owner arn:aws:sts:851725596827:assumed-role/voclabs/user4731589-Abbas_Muzan
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Launch template version details Actions ▼ Delete template version

Version: 1 (Default) Description: v1 Date created: 2025-04-03T15:49:49.000Z Created by: arn:aws:sts:851725596827:assumed-role/voclabs/user4731589-Abbas_Muzan

Instance details Storage Resource tags Network interfaces Advanced details

AMI ID ami-01b14b7ad41e17ba4	Instance type t2.micro	Availability Zone -	Availability Zone Id -
Key pair name keypem	Security groups -	Security group IDs sg-00b37e48fafead3c7	

2) Auto Scaling Group Desired: 2, Min:2, Max:6

Muzan-ASG

Muzan-ASG Capacity overview

arn:aws:autoscaling:us-east-1:851725596827:autoScalingGroup:7ea3186e-6e7e-4b22-bbac-8c9f992f972d:autoScalingGroupName/Muzan-ASG

Desired capacity

2

Scaling limits

2 - 6

Desired capacity type

Units (number of instances)

Status

-

Date created

Sun Apr 05 2026 13:42:34 GMT+0300 (Arabian Standard Time)

Details

Integrations

Automatic scaling

Instance management

Instance refresh

Activity

Monitoring

Tags

Launch template

Launch template

lt-0ddafcdc07656d5c5

WebTier-LT

Version

Latest

Description

-

AMI ID

ami-01b14b7ad41e17ba4

Security groups

-

Storage (volumes)

-

Instance type

t2.micro

Security group IDs

sg-00b37e4f6afead3c7

Key pair name

keyperm

Owner

arn:aws:sts::851725596827:assumed-role/voclabs/user4731589-Abbas_Muzan

Create time

Mon Apr 06 2026 22:21:10 GMT+0300 (Arabian Standard Time)

Request Spot Instances

No

Network

Availability Zones

use1-az1 (us-east-1b)
use1-az6 (us-east-1a)

Subnet ID

subnet-0186a826ab12fac46
subnet-04106b5173f16579a

Availability Zone distribution

Balanced best effort

Instance type requirements

Your Auto Scaling group adheres to the launch template for purchase option and instance type.

Muzan-ASG

Muzan-ASG Capacity overview

arn:aws:autoscaling:us-east-1:851725596827:autoScalingGroup:7ea3186e-6e7e-4b22-bbac-8c9f992f972d:autoScalingGroupName/Muzan-ASG

Desired capacity

2

Scaling limits

2 - 6

Desired capacity type

Units (number of instances)

Status

-

Date created

Sun Apr 05 2026 13:42:34 GMT+0300 (Arabian Standard Time)

Details

Integrations

Automatic scaling

Instance management

Instance refresh

Activity

Monitoring

Tags

Launch template

Launch template

lt-0ddafcdc07656d5c5

WebTier-LT

Version

Latest

Description

-

AMI ID

ami-01b14b7ad41e17ba4

Security groups

-

Storage (volumes)

-

Instance type

t2.micro

Security group IDs

sg-00b37e4f6afead3c7

Key pair name

keyperm

Owner

arn:aws:sts::851725596827:assumed-role/voclabs/user4731589-Abbas_Muzan

Create time

Mon Apr 06 2026 22:21:10 GMT+0300 (Arabian Standard Time)

Request Spot Instances

No

Network

Availability Zones

use1-az1 (us-east-1b)
use1-az6 (us-east-1a)

Subnet ID

subnet-0186a826ab12fac46
subnet-04106b5173f16579a

Availability Zone distribution

Balanced best effort

Instance type requirements

Your Auto Scaling group adheres to the launch template for purchase option and instance type.

Group size

Specify the size of the Auto Scaling group by changing the desired capacity. You can also specify minimum and maximum scaling limits.

Desired capacity type

Choose the unit of measurement for the desired capacity value. vCPUs and Memory(GiB) are only supported for mixed instances groups configured with a set of instance attributes.

Units (number of instances)

Desired capacity

Specify your group size.

2

Scaling limits

Set limits on how much your desired capacity can be increased or decreased.

Min desired capacity

2

Equal or less than desired capacity

Max desired capacity

6

Equal or greater than desired capacity

Cancel

Update

3) 2 Private Subnets

Instance type requirements

Override launch template

You can keep the same instance attributes or instance type from your launch template, or you can choose to override the launch template by specifying different instance attributes or manually adding instance types.

Launch template

WebTier-LT

lt-0ddafcd07656d5c5

Version

Latest

Description

Instance type

t2.micro

Network

For most applications, you can use multiple Availability Zones and let EC2 Auto Scaling balance your instances across the zones. The default VPC and default subnets are suitable for getting started quickly.

Availability Zones and subnets

Define which Availability Zones and subnets your Auto Scaling group can use in the chosen VPC.

Select Availability Zones and subnets

us-east-1a | subnet-04106b5173f16579a (Private-Subnet-1)

10.0.1.0/24

us-east-1b | subnet-0186a826ab12fac46 (Private-Subnet-2)

10.0.3.0/24

Create a subnet

Availability Zone distribution - new

Auto Scaling automatically balances instances across Availability Zones. If launch failures occur in a zone, select a strategy.

Balanced best effort

If launches fail in one Availability Zone, Auto Scaling will attempt to launch in another healthy Availability Zone.

Balanced only

If launches fail in one Availability Zone, Auto Scaling will continue to attempt to launch in the unhealthy Availability Zone to preserve balanced distribution.

Cancel

Update

4) Scaling Policy with CPU target tracking at 60%

Muzan-ASG Capacity overview

Edit

arn:aws:autoscaling:us-east-1:851725596827:autoScalingGroup:7ea3186e-6e7e-4b22-bbac-8c9f992f972d:autoScalingGroupName/Muzan-ASG

Desired capacity

2

Scaling limits

2 - 6

Desired capacity type

Units (number of instances)

Status

-

Date created

Sun Apr 05 2026 13:42:34 GMT+0300 (Arabian Standard Time)

Details

Integrations

Automatic scaling

Instance management

Instance refresh

Activity

Monitoring

Tags

Scaling policies resize your Auto Scaling group to meet changes in demand. With reactive dynamic scaling policies, you can track specific CloudWatch metrics and take action when the CloudWatch alarm threshold is met. Use predictive scaling policies along with dynamic scaling policies in the following situations: when your application demand changes quickly, but with a recurring pattern, or when your EC2 instances require more time to initialize.

Dynamic scaling policies (1)

Info

Actions

Create dynamic scaling policy

< 1 >

Target Tracking Policy

Policy type

Target tracking scaling

Enabled or disabled

Enabled

Execute policy when

As required to maintain Average CPU utilization at 60

Take the action

Add or remove capacity units as required

Instances need

300 seconds to warm up before including in metric

Scale in

Enabled

5) CloudWatch Alarm

CloudWatch

Alarms Mute rules - new

Favorites and recents

Ingestion

Dashboards

Alarms 1 0 0

AI Operations

GenAI Observability

Application Signals (APM)

Infrastructure Monitoring

Logs

Metrics

Network Monitoring

Setup

Alarms (2)

Clear selection

Create composite alarm

Actions

Create alarm

Filter alarms

Quick filters

☐	Name	State	Actions	Actions muted - new	Last state update (UTC)	Conditions
☐	TargetTracking-Muzan-ASG-AlarmHigh-c29e0f66-7f8c-4754-89ef-88e40e6f8814	OK	Actions enabled	-	2026-04-06 16:29:34	CPUUtilization > 60 for 3 datapoints within 5 minutes
☐	TargetTracking-Muzan-ASG-AlarmLow-bbe477bc-0a1d-4b07-88e4-1cd118297194	In alarm	Actions enabled	-	2026-04-06 16:28:55	CPUUtilization < 42 for 15 datapoints within 15 minutes

Conclusion:

Overall, I was able to successfully build the AWS architecture based on the given requirements. The system is designed to be secure, scalable, and highly available by using multiple Availability Zones, separating public and private resources, and controlling traffic between different layers. All components were configured to match the required setup, and the final architecture works as intended.